

# Evaluating the antimicrobial activity and antitumor screening of green synthesized silver nanoparticles compounds, using *Syzygium jambolanum*, towards MCF7 cell line (Breast cancer cell line)

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## ABSTRACT

In this study, rapid, simple approach was applied for synthesis of silver nanoparticles using *S. jambolanum* aqueous leaf extract. The plant extract acts both as a reducing agent as well as capping agent. To identify the compounds responsible for reduction of silver ions, the functional groups present in plant extract were investigated by FTIR. Various techniques used to characterize synthesized nanoparticles are UV Visible spectrophotometer, Scanning Electron Microscopy (SEM), Energy dispersive X-ray analyser (EDX). UV-Visible spectrophotometer showed absorbance peak in the range of 335 nm. The NPs displayed higher antioxidant activity, antibacterial activities against both gram positive and gram negative bacteria. These NPs also exhibited effective anti-cancerous activity with a median inhibitory concentration (IC<sub>50</sub>) of 0.2588 ppm/100μL against MCF-7 breast cancer cell line in cytotoxic assay and indicated antihaemolytic activity at less concentration. Results confirmed this protocol as simple, rapid, one step, eco-friendly, nontoxic and an alternative conventional physical/chemical method. Only 15 min were required for the conversion of silver ions into silver nanoparticles at room temperature, without the involvement of any hazardous chemical.

## 1. Introduction

Nanotechnology is a key area dealing with the manufacture, management and use of the assets ranging in nanometers. Nanotechnology offers fields with effective applications, ranging from traditional chemical techniques to medicinal and environmental technologies. Ag-NPs have emerged with leading contributions in diverse applications as nanomedicine [59]. Despite the contradictions reported on the toxicity of AgNPs, its role as a disinfectant and antimicrobial agent has been given considerable appreciation [20]. The development of nanoparticles containing metal oxide may result in environmental contaminations, since the chemical procedures involved in the synthesis of nanomaterials generate many hazardous by products. In this sense, the green synthesis aims at synthesizing nanoparticles from plant extracts, bacteria, fungal species, and others. The synthesis of nanoparticles does not require high energy or temperature, and it is easily scaled up for large-scale synthesis. This is also considered to be cost effective option. This technique employs natural plant biology or a vegetation source to synthesize nanoparticles. This 'green' eco-friendly method using chemistry

and chemical technology has become extremely famous and useful due to rising worldwide concerns associated with environmental issues [42].

Silver [Ag] is a white, lustrous transition metal with high electrical and thermal conductivity quite useful in the manufacture of foils, vessels and highly sensitive biosensors for biomolecular detection. It's bactericidal, anti-fungal, anti-inflammatory and anti-angiogenesis properties make it a potent healing agent for use as medicine against highly infectious diseases and surgical infections [38,47]. These advantages of silver make it a better choice for producing nanoparticles over other elements.

Green synthesis of nanoparticles will help in the development of strategies that are simpler, profitable, environment friendly and relatively easy to reproduce resulting in manufacture of more stable substances [12,21]. Plant mediated AgNPs can also be produced from legume flowers grown in the Indian subcontinent. Although the use of flora as a potential source for nano-particle fabrication is still largely unexplored, researchers have used extracts from *Ziziphora tenuior*, *Abutilon indicum*, *Solanum tricobatum*, *Erythrina indica*, *Ocimum tenuiflorum*, *Spirogyra varians*, *Meliadubia*, *Acalypha indica* leaf extract and *Sesuvium*

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Fig. 1. Image of *Syzygium jambolanum*.

*portulacastrum* to fabricate NPs with varying lengths mostly in the range 5 – 20 nm [2,4–7,10,22,25,33,34,39,40,48,49,58,61].

The flowering plant *Syzygium jambolanum* belongs to the family of myrtaceae native to the India. The leaves have an aroma like turpentine and the fruit seed contains vitamin A and C in high amounts, used in Ayurveda and Chinese medicine for controlling diabetes or digestive illness [1,69]. Silver nanoparticles [AgNPs] can be produced even with low amounts of leaf extract without the use of any dangerous chemical or physical procedures. The effect of concentration of silver ions and leaf extract amounts were additionally evaluated to optimize the process for silver nanoparticle synthesis. This approach is clean and easy to perform along with being cost effective and sustainable.

Breast carcinoma is the most common prevalent malignancy in women [44] and it is the cause for morbidity and mortality for women worldwide [27,68]. A variety of approaches are being practiced for the treatment of cancer each of which has some significant limitations and side effects [3]. Conventional chemotherapy fails to target the cancerous cells particularly without interacting with the normal body cells. Hence, there is a need to avoid the toxic chemicals in order to circumvent adverse effects in medical applications. Therefore, the present study aims at synthesizing the AgNPs using *S. jambolanum* leaves extracts can be useful and their effect on breast cancer cell lines. Hence this is green synthesis of makes an alternative to treat cancer to avoid problems concerned with conventional chemotherapy.

## 2. Materials and methods

### 2.1. Study material

Fresh leaves of *Syzygium jambolanum* collected from Botanical garden (14°28'25.21" and 78°42'44.36"E), Yogi Vemana University, Kadapa (A.P), India. (Fig. 1)

### 2.2. Preparation of *S. jambolanum* nanoparticles

#### 2.2.1. *S. jambolanum* leaf extract preparation

The young leaves of *S. jambolanum* were collected and washed with tap water to remove debris. The leaves were then dried using blotting papers. The leaves were dried before being finely ground using a mortar and pestle [23].

#### 2.2.3. Preparation of 1 mM silver nitrate [AgNO<sub>3</sub>] solution

For the preparation of the 1 mM solution, 0.017gms of AgNO<sub>3</sub> was added to 100 ml of double distilled water. The combined solution was transferred to a dark amber bottle preventing any oxidation of silver [Ag] [29].

#### 2.2.4. Green synthesis of *S. jambolanum* Ag NPs

The synthesis of plant mediated Ag NPs involves mixing the leaf extract and 1 mM AgNO<sub>3</sub> solution in 1:4 ratio. The mixture is stirred constantly on a hot plate at 70–80°C for 30 min or until the solution color changes to light brownish color [13].

### 2.3. Characterization of *S. jambolanum* silver nanoparticles

#### 2.3.1. Determination of absorption maxima of *S. jambolanum* AgNPs by UV-Vis spectra analysis

UV-Vis spectrophotometer (ELICO SL-159) monitors the of pure Ag<sup>+</sup> ions reduction. 1 mL of AgNPs sample was taken into a quartz cuvette and introduced into UV-Vis spectrum was analyze the response after 30 min in range of 350–750 nm [57].

#### 2.3.2. FTIR analysis

In order to prevent contamination of the capping ligand of Nanoparticles by biomass residue, 100 mL of the residual solution was centrifuged at 15,000 rpm for 10 min after the reaction was complete. The resulting suspension was washed with sterile distilled water and purified. The suspension was then dried to obtain a stable powder containing the dried nanoparticles which were subject to FTIR analysis using FTIR-8400S Shimadzu [63] to determine the functional groups.

#### 2.3.3. SEM - EDX analysis

Scanning Electron Microscopic (SEM) analysis was accomplished with the use of Hitachi S - 4500 SEM machine. Thin films of synthesized and stabilized AgNPs were prepared on a carbon coated copper grid and the pattern was analyzed for morphology and size of the nanoparticles [46].

#### 2.3.4. Determination of particle size and zeta potential of *S. jambolanum* AgNPs by particle size analyzer and zeta potential analyzer

Synthesized nanoparticles size and zeta potential were measured by zeta sizer nanoseries between 0.1 to 10,000 nm (Nanopartica SZ-Horiba, Japan). Stability of the nanoparticle sample was measured by maintaining the temperature of 25 °C dispersion, 0.894 mPa of dispersion medium viscosity, 0.073 mS/cm conductivity with 3.8 V electrode voltages.

### 2.4. Biological characterization of *S. jambolanum* AgNPs

#### 2.4.1. Phytochemical screening

By adopting standard procedure [28,35,36], the sample extracts from each plant were utilized to screen in order to identify their phytochemical contents.

#### 2.4.2. Invitro antioxidant assay

DPPH2 (2- Diphenyl-1-picrylhydrazyl) consists of free nitrogen free radical. In the presence of proton radical scavenger/hydrogen donating antioxidants, DPPH was converted into a stable DPPH-H. The quenching activity of free radicals was determined for *S. jambolanum* AgNPs from

5–25  $\mu\text{gml}^{-1}$ . Finalize the volume with 250  $\mu\text{l}$  of DMSO with addition of 1.5 ml of 0.1 mM DPPH followed by vortexing, kept this in dark condition at room temperature for 30 min. The absorbance was recorded at 517 nm. Standard graph was plotted by using BTH as standard to give inhibition curve along with the test sample [17]. The free radical scavenging activity was calculated by the following equation;

$$\text{Percentage of inhibition} = \left[ \frac{(\text{Standard OD} - \text{Test OD})}{\text{Standard OD}} \right] \times 100$$

All the biological experiments were conducted five times and the statistical analysis was carried out by Tukey's Multiple Comparison Test from one-way ANOVA test ( $p < 0.001$ ) to interpret the significant differences.

#### 2.4.3. Antibacterial activity

*S. jambolanum* AgNPs antibacterial activity was assessed against *Escherichia coli*, *Staphylococcus aureus*, and *Pseudomonas aeruginosa*, using the disk diffusion approach. Whatmann filter paper No.1 discs were prepared and soaked in *S. jambolanum* AgNPs solution and then air dried in sterile conditions. New agar medium plates were prepared by swabbing with microbial cultures and divided into 4 identical parts with each part being placed upon the previously dried discs. The plates were then incubated at 37 °C for 24 – 48 h and observed for zone of bacterial inhibition [56,65]. All the biological experiments were conducted five times and the statistical analysis was carried out by Tukey's Multiple Comparison Test from one-way ANOVA test ( $p < 0.001$ ) to interpret the significant differences.

#### 2.4.4. Cell culture

MCF-7 (Breast cancer cell line) were obtained from National Centre for Cell Science, Pune in Maharashtra, India and maintained on RPMI – 1640 media supplemented with 10% v/v fetal bovine serum, sodium carbonate ( $\text{Na}_2\text{CO}_3$ ) along with 100  $\text{mg L}^{-1}$  penicillin and 50  $\text{mg L}^{-1}$  streptomycin to prevent any bacterial contamination with incubation at 37 °C and 5%  $\text{CO}_2$  in a humidified atmosphere.

**Anticancer assay:** MTT (3–4, 5– dimethylthiazol–2–yl) – 2, 5–diphenyl tetrazolium bromide) assay was performed to know the cytotoxicity of *S. jambolanum* AgNPs. Microplate reader was used to perform MTT assay at 560 nm to calculate the percentage of cells survival and to determine the concentration of Ag NPs required to kill 50% ( $\text{IC}_{50}$ ) of the cancer cells compared to that of the control. Percentage of growth inhibition (GI%) was measured by plotting a bar graph with concentration of synthesized compounds on X-axis and cell viability (%) on Y – axis. (GI%) [26]. The above experiments were conducted five times and the statistical analysis was carried out by Multiple Comparison Test from two-way ANOVA test ( $p < 0.001$ ) to interpret the significant differences.

**In-vitro haemolytic activity:** Human blood sample was collected with anticoagulant solution (Alsever's solution) and in the sterile container followed by centrifugation at 1500 rpm for 10 min at room temperature to collect RBC's. The RBC solution was prepared by adding 5 mL PBS to remove heme into the obtained pellet. Four different concentrations of extract namely 22, 70, 220, 700 and 2200  $\mu\text{g/ml}$  of *S. jambolanum* AgNPs. To the reaction mixture 0.5 ml of RBC solution was added and makes the volume to 1 mL by PBS, with incubation at 37 °C for 30 min followed by centrifugation at 1500 rpm for 5 min and absorbance was measured at 540 nm. Triton-X 100 (0.1%) was used as a positive control and RBC without control was used as negative control [62].

% of haemolysis =

$$\left[ \frac{(\text{Abs. of Test sample} - \text{Abs. of Negative control})}{(\text{Abs. of positive control} - \text{Abs. of Negative control})} \right] \times 100.$$

The above experiments were conducted five times and the statistical analysis was carried out by Multiple Comparison test from two-way ANOVA test ( $p < 0.001$ ) to interpret the significant differences.

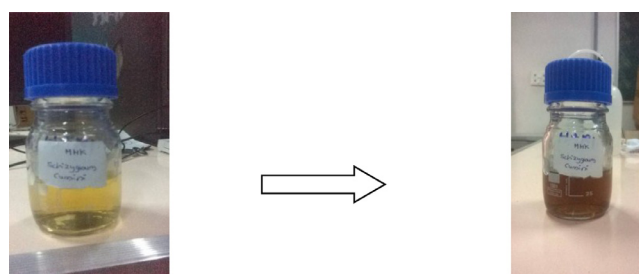


Fig. 2. color change indicates the formation of silver nano particles.

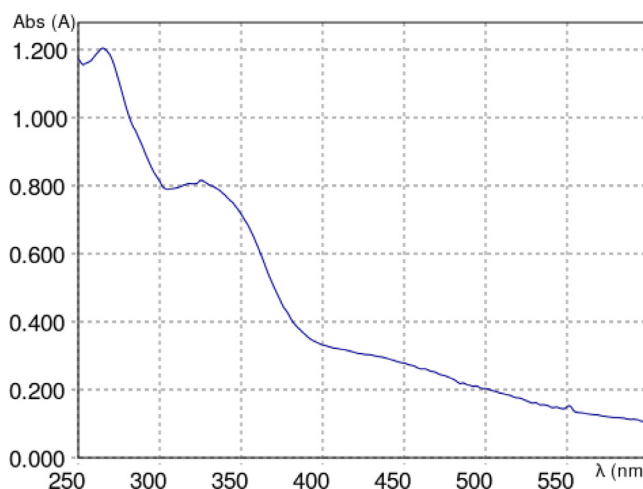


Fig. 3. UV-VIS spectrophotometry peak maximum at 335 nm.

### 3. Results and discussion

#### 3.1. Characterisation of *S. jambolanum* AgNPs

##### 3.1.1. UV-Vis spectroscopy analysis of *S. jambolanum* AgNPs

*S. jambolanum* leaf extract was added to an aqueous silver nitrate solution and incubated at room temperature for 30 min. After incubation the solution was changed from colourless to brownish gray results in the formation of nanoparticles (Fig. 2). Brownish gray color of silver nanoparticles is due to excitation of surface plasmon vibrations. The absorption maxima of *S. jambolanum* nanoparticles was 335 nm (Fig. 3). The UV-Vis spectrum was recorded in time intervals of 30 min for 24 h from the initiation of the reaction. AgNPs give absorption peak at 310 and 395 nm and due to surface plasmon resonance (SPR) reported earlier [43].

##### 3.1.2. FTIR analysis

FTIR spectra of the biosynthesized silver nanoparticles and the aqueous plant extract are shown in Fig. 4a and 4b. Responsible metabolites for reduction of Ag ions were detected using FTIR spectrum. Instinct absorption bands at 3443  $\text{cm}^{-1}$  appear in the presence of amine N–H in the nanoparticles whereas 3625  $\text{cm}^{-1}$  in the amide N–H stretch in the plant extract. The plant extract was superimposed by 3299, 3825  $\text{cm}^{-1}$  in the amide N–H stretch in the plant extract. 2925  $\text{cm}^{-1}$  corresponds to Alkyl C–H group in the AgNP's, whereas 2855  $\text{cm}^{-1}$ , 2938  $\text{cm}^{-1}$ , 2835  $\text{cm}^{-1}$  in the plant extract also corresponds to Alkyl C–H group in the plant extract. 2519  $\text{cm}^{-1}$  corresponds to carboxylic O–H stretch in the AgNPs whereas, 2321  $\text{cm}^{-1}$  corresponds to carboxylic O–H stretch in the plant extract. 1631  $\text{cm}^{-1}$  corresponds to Amide C=O stretch in the AgNPs whereas, 1696  $\text{cm}^{-1}$ , 1622  $\text{cm}^{-1}$  corresponds aromatic c=c bending in the plant extract. Only minor changes in the position of absorption bands between FTIR spectrum of the plant extract and biosynthesized Ag NPs were seen. The involvement of plant extract compounds in



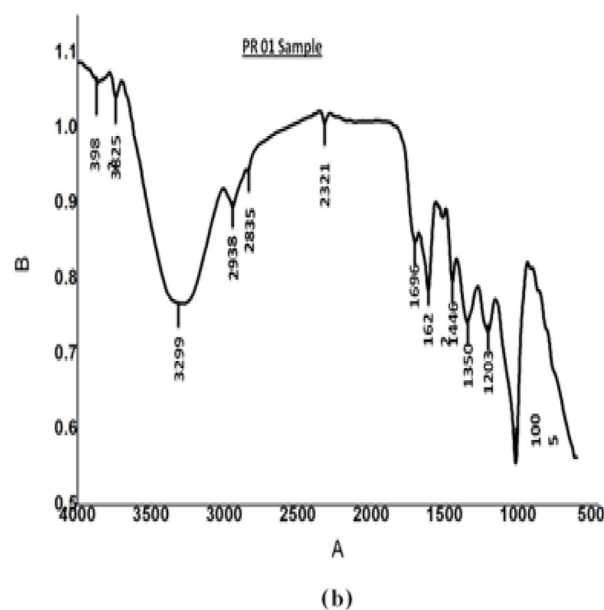
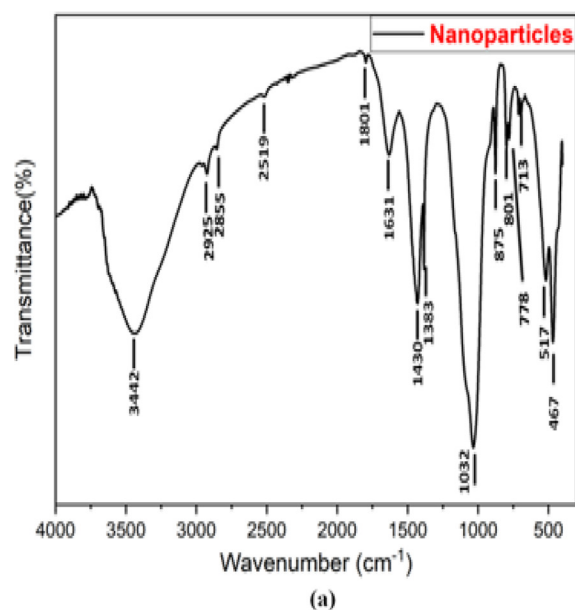


Fig. 4. a & 4b. FTIR spectrum of Ag NPs and plant extract.

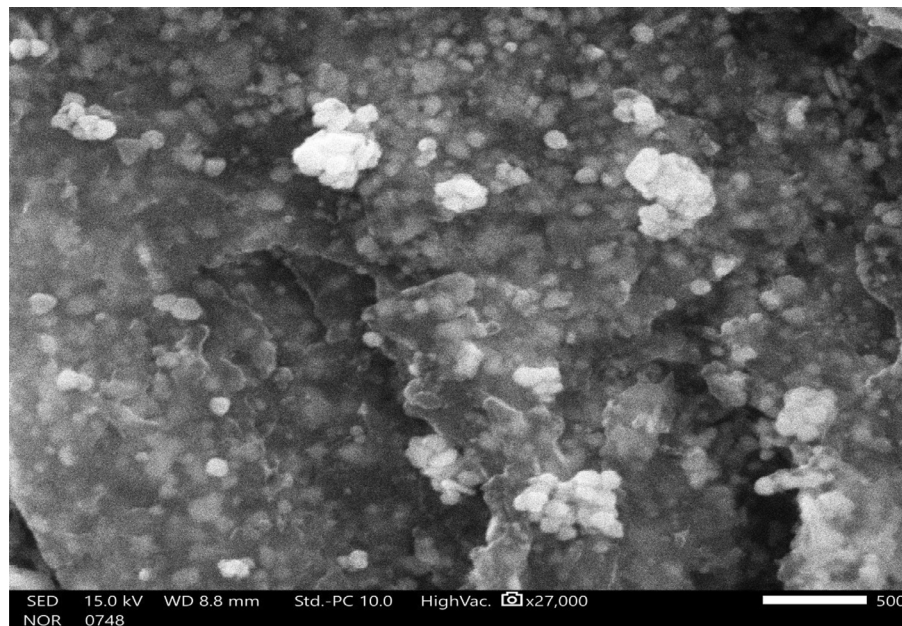


Fig. 5. SEM analysis of silver nanoparticles.

biosynthesizing of nanoparticles was confirmed with shifting of peaks. Thus, plant extract compounds including OH groups have a vital role in reducing and stabilization of NPs. The results from FTIR analysis concluded that the bio-organic compounds from *S. jambolanum* leaf extract forms a strong coating/capping on the NP's reported by other researchers [30,51,53].

### 3.1.3. SEM analysis

The SEM images of *S. jambolanum* of AgNPs are provided in Fig. 5 and shows the spherical shapes obtained as a result of the different leaf extracts being used as reducing and capping agents. The green synthesis of *S. jambolanum* leaf extract nanoparticles SEM analysis shows that nanoparticles were well-defined with distinctly ordered and skinny sheets. The majority of the thin AgNPs sheets showed interspersed association with each other over a large surface area. This could be due to the different amount and nature of capping agents present within the

leaf extracts [14,45,51,64,66]. Further Kadam et al. [32] revealed that cauliflower waste – AgNPs SEM analysis showed nanoparticles ranging in size between 20.2 to 24.5 nm.

### 3.1.4. EDX analysis

The peaks corresponding to Ag were assigned. The peaks for Oxygen (O) and Carbon (C) were used from the grid while Sulphur (S), and Nitrogen (N) peaks corresponding to the proteins include capping on the AgNPs (Fig. 6). The Ag nanocrystallites showed maximum at (3 keV) X-ray fluorescence [8,14,64].

### 3.1.5. DLS and zeta potential analysis of *S. jambolanum* - AgNPs

Hydrodynamic size and surface charge of AgNPs were revealed by DLS. The AgNP's showed average diameter of 8.3 nm which confirm its polydisperse in nature and their stability was determined as  $-52.5$  mv (Fig. 7a and 7b) with particle size and zeta potential analyzer. In this

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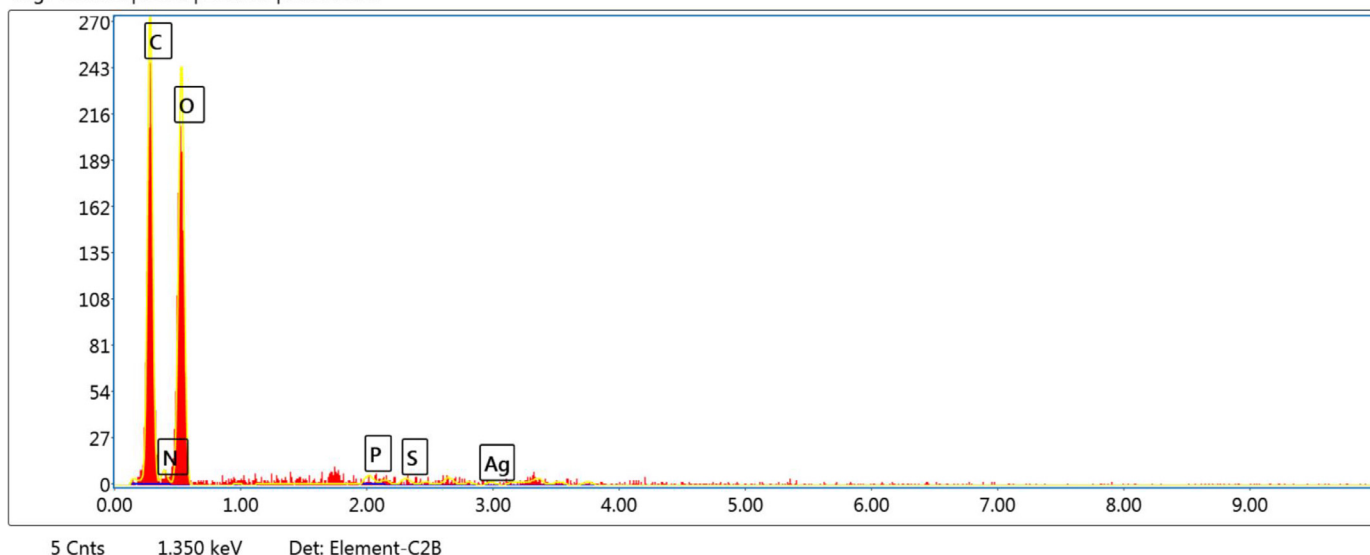
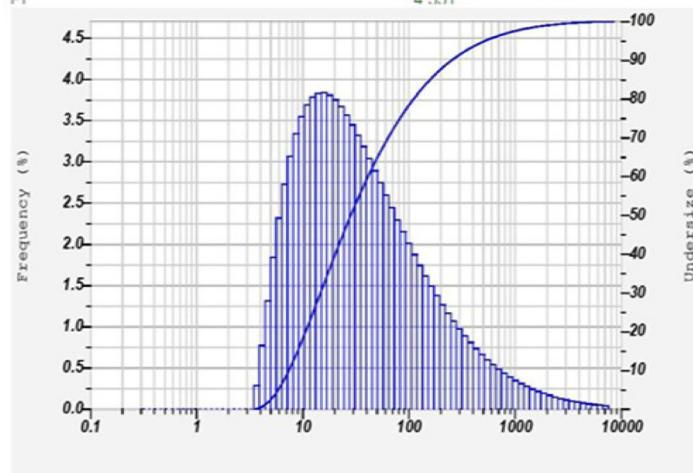


Fig. 6. EDX spectrum of synthesized silver nanoparticles.

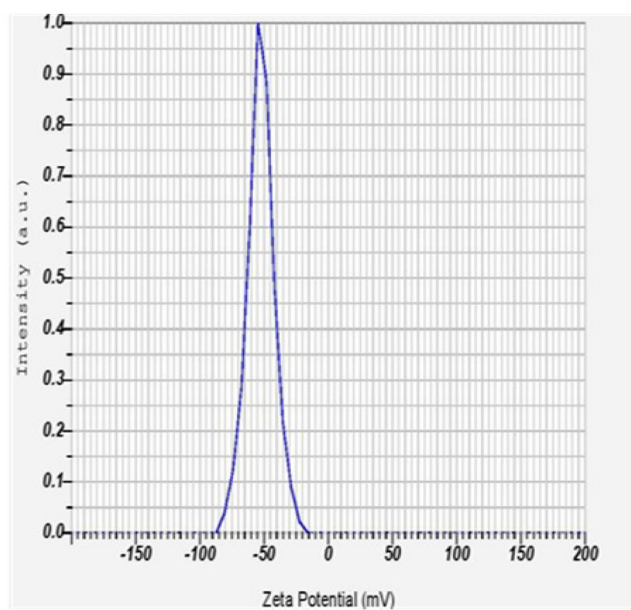
## Calculation Results

| Peak No. | S.P Area Ratio | Mean     | S. D.    | Mode    |
|----------|----------------|----------|----------|---------|
| 1        | 1.00           | 131.7 nm | 424.1 nm | 15.8 nm |
| 2        | —              | — nm     | — nm     | — nm    |
| 3        | —              | — nm     | — nm     | — nm    |
| Total    | 1.00           | 131.7 nm | 424.1 nm | 15.8 nm |

## Cumulant Operations

Z-Average  
PI: 8.3 nm  
: 4.351

(a)



(b)

Fig. 7. Particle size and Zeta potential of nanoparticles.

connection, similar report from Deepika et al. [16] showed that the average diameter of the synthesized AgNP's was 155.4 nm at  $-23.4$  mV. Vorobyova et al. [67], that spherical AgNPs of currant pomace extract displayed 40–60 nm particle size with a polydispersity (PDI) of 0.26–0.78 and with moderate stability (zeta potential of  $-13.41$  mV). Saratale et al. [60] found, that zeta potential of *Acacia nilotica* leaves AgNPs displayed higher negative charge  $-27.75$  mV. This negative charge of NPs is suggestive of repelling of particles with each other, thereby avoiding agglomeration. Owing to their long-term stability, the synthesized *Acacia nilotica* leaves AgNPs seems to be appropriate in different biomedical sectors. Pilaquinga et al. [50] observed that *Mimosa albida* AgNPs mean value obtained by DLS was  $9.3 \pm 0.3$  nm. Jebriil and Dridi, [23] reported that *M. azedarach* leaf extract AgNPs shows negative zeta potential value of  $-13.1$  mV indicated the stability and good dispersion of AgNPs.

3.2. Biological characterization of *S. jambolanum* AgNPs

## 3.2.1. Phytochemical screening

*S. jambolanum* leaf extract was subjected qualitatively for phytochemical screening (Table 1). The analysis shows a high level of Phenolic and phenolic compounds [54]

## 3.2.2. Invitro antioxidant assay

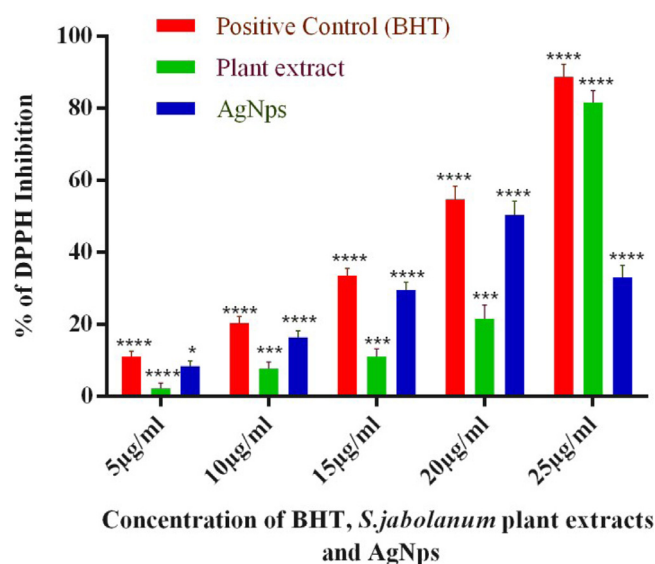
DPPH assay was performed and the results are depicted in Fig. 8. Inhibition of free radicals was noted with an increase in extracts concentration. The *S. jambolanum* AgNPs was observed to have a more effective scavenging activity of free radicals compared to the plant extract. At 25  $\mu\text{g/ml}$  concentration of AgNPs shows 81% of free radical scavenging activity. The defense mechanism of all organisms can reduce, scavenge

**Table 1**  
Phytochemical analysis of *Syzygium jabolanium*.

| S. No | Test               | Result   |
|-------|--------------------|----------|
| 1.    | Mayer's test       | (-)      |
| 2.    | Benedict's test    | (+++)    |
| 3.    | Saponins           | (++++++) |
| 4.    | Phenolic compounds | (++++++) |
| 5.    | Tannins            | (++)     |
| 6.    | Flavonoids         | (+)      |
| 7.    | Terpenoids         | (++)     |

(+)=presence.

(-)=absence.



**Fig. 8.** DPPH scavenging activity of *S. jabolanium* leaf extracts and their AgNP's along with Positive control (BHT). Data was measured in at least 5 samples, S.D. are shown. Statistical analysis was done using TWO WAY ANOVA with  $P < 0.001$ . \*Denotes significant difference of positive control with test samples at  $P < 0.001$ .

and stabilize the production of free radicals (hydroxyl, superoxide, nitric oxide and lipid peroxy radicals) and non-free radical species ( $H_2O_2$  and singlet oxygen).

The above results were similar to the *Acacia nilotica* leaves - AgNPs exhibited noteworthy DPPH free radical scavenging ability having IC<sub>50</sub> value of 55.2 µg/mL almost quite equivalent with the standard catechol [60]. In this connection, Pilaquinga et al. [50] reported the *Mimosa albida* AgNP's antioxidant was maximum determined through a Trolox calibration curve. In similar manner, Vorobyova et al. [67] revealed that the nanoparticles (Black Currant Pomace Extract-AgNPs) showed higher antioxidant activity than the Black Currant Pomace Extract. Further, Jini and Sharmila [24] reported that the synthesized silver nanoparticles from *Allium cepa* showed a better DPPH scavenging activity and it was dose dependent, it showed that the synthesized silver nanoparticles have good antioxidant activity which may be due to the presence of functional groups from the *Allium cepa* extract in the synthesized silver nanoparticles.

### 3.2.3. Antimicrobial activity

Antimicrobial activity of *S. jabolanium* plant extract and their nanoparticles were shown in Fig. 9. *S. jabolanium* nanoparticles showed the greatest zone of inhibition against *P. aeruginosa* (Fig. 9(c)) followed by *E. coli* (Fig. 9(b)) and *S. aureus* (Fig. 9(a)). Since the plant leaf is presumed to possess most antibacterial properties, it is expected that *S. jabolanium* leaf mediated green synthesis should have greater zone of inhibition compared to the plant extract zone of inhibition. The AgNPs

showed better antimicrobial properties as surface area is larger offering better contact with cellular wall of microorganisms.

The above results were similar to *J. procera* plant extract AgNPs shows more activity both Gram positive and negative bacteria as well as yeast [31]. Further, Pirtarighat et al. [51] noticed that the *Salvia spinosa* AgNPs the inhibit the growth of bacteria of *B. subtilis*, *B. valislimortis* and *E. coli*. Similar evidences were provided by Mariadoss et al. [37] that the *Malus domestica*-AgNPs revealed the dose-dependent antimicrobial activity against *E. aerogenes*, *K. pneumoniae*, *P.aeruginosa*, and *E.coli*. Awwad et al. [9] observed that an increase in rosemary - Ag<sub>2</sub>SNPs concentration increases the maximum zone of Inhibition of *E. coli*, *S. aureus*, *Shigella* and *Listeria* bacteria. Banasiuk et al. [11] revealed the *D. muscipula* and *D. Indica* AgNPs at 5.3 mg/mL and 42.5 mg/mL. The most resistant to AgNPs was fungi (*C. albicans*) where MBC was above 170 mg/ml.

### 3.2.4. Cytotoxicity evaluation

*S. jabolanium* AgNPs cell viability was determined by MTT assay. It was observed that AgNPs suppressed the growth of cancer cells (Fig. 10) showing significant growth inhibition of MCF - cell line, at all concentration. The results indicated that *S. Jabolanium* AgNPs caused a significant decrease in cell viability of the MCF-7 cells when compared to the plant extract. In the study of Mariadoss et al. [37], the results of experiment using cytotoxicity impact on cell growth was observed different concentration of (10–100 µg/ml), the IC<sub>50</sub> value of *Mallus domestica* - AgNPs observed at the concentration of 33.81 µg/ml against MCF-7 cells. [31] reported that the maximal stimulation was at higher concentration of *Juniperus procera* containing nanoparticles (200 mg/mL) on PHA stimulated spleen cells. In this connection, the *Ananas comosus*- AgNPs expressed high anti-diabetic potential with higher cytotoxic activity against the HepG2 cancer cell lines in a dose-dependent manner [15]. Similar evidence was provided by Erdogan et al. [19] with combined treatment with 10 µg/mL *Cynara scolymus* leaf extracts AgNPs+PDT therapy in the MCF7 cells suppressed cell proliferation after 24 h. Further Hamouda et al. [26] noticed that the *Oscillatoria* - AgNPs shows cytotoxic effect with (IC<sub>50</sub>) was observed at 6.147 µg/ml against MCF-7 and 5.369 µg/ml for HCT-116 cells.

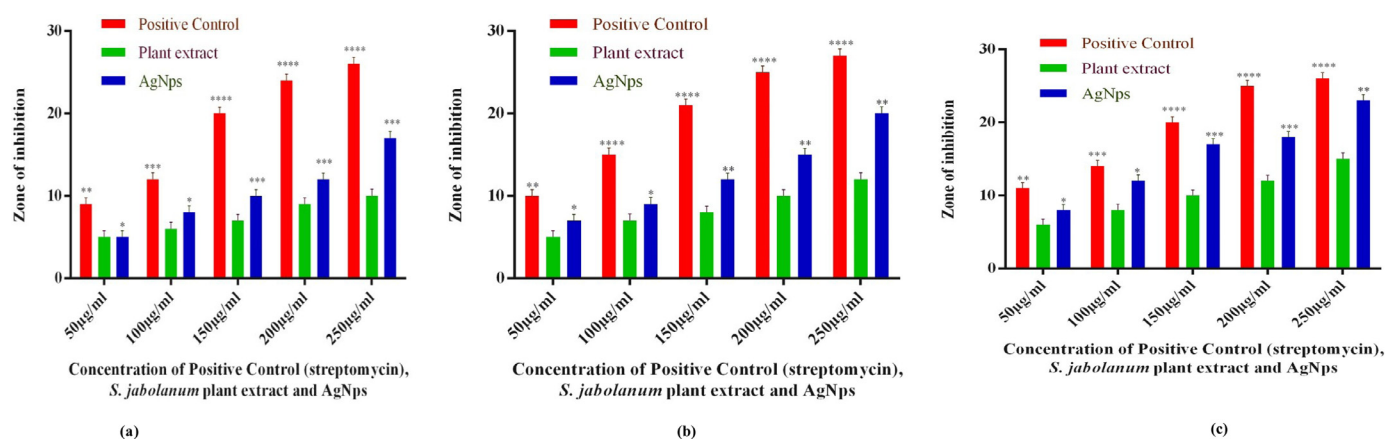
### 3.2.5. Invitro haemolytic activity

Biological substance with hemolytic activity of greater than 5% are known as hemolytic, between 2 and 5% are slightly hemolytic, while less than 2% is non-hemolytic [18]. If a given nanoparticle is hemolytic, it will rupture red blood cells, which further result in hemoglobin release. To confirm the bio-safe nature, hemolysis assay was conducted using human RBCs. *S. jabolanium* plant extract and their silver nanoparticles shows the least haemolytic activity at 22 µg/100 µl, 70 µg/100 µl. 1.2% and 1.5% of haemolysis at 22 µg/100 µl and at 70 µg/100 µl 2.2% and 2.3% (Fig. 11) by plant extract and AgNps respectively. This indicates that these nanoparticles and plant extracts are non-haemolytic making its nontoxic.

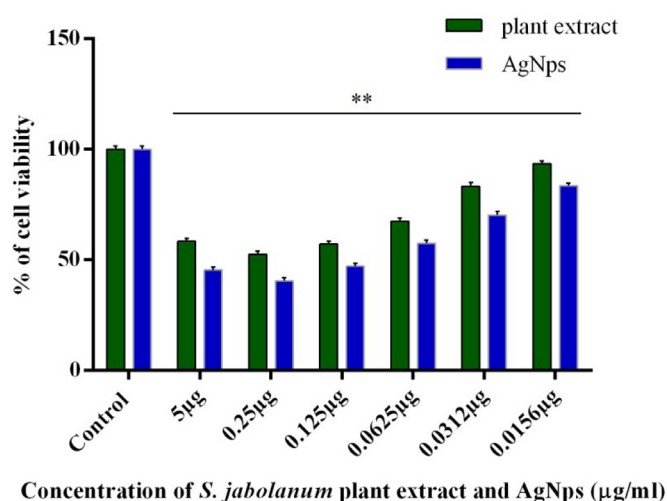
But at higher concentrations both plant extract and AgNps showed haemolytic activity indicating that these nanoparticles are slightly haemolytic in nature (Fig. 11). But increasing concentration of the nanoparticles increase in haemolysis activity. Hence the RBC cell membrane integrity was maintained at less concentration. Hence the *S. jabolana* silver nanoparticles extract can be considered for pre-clinical and clinical studies.

The above results are similar with [55] reported that the synthesis of AgNPs from *Annona reticulata* leaves extracts on percentage of lysis was obtained only about  $13.8 \pm 1.72$  to  $0.87 \pm 0.28$  at 500 to 31.25 mg/ml concentration respectively. Hamouda et al. [26] obtained an induced hemolytic action on the erythrocytes as a function of increasing *Oscillatoria*-AgNPs concentration. At 2.5, 5.0, 7.5, 10 µg/ml O-AgNPs concentrations shows 0.45, 0.76, 1.03, 1.68% of haemolysis respectively. Preliminary toxicity was observed in the study of various concentration of *Eucalyptus citriodora* AgNPs on human red blood

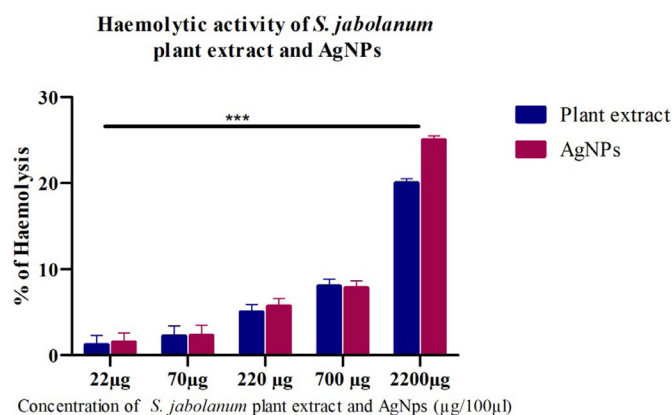




**Fig. 9.** Antibacterial activity of *S. jabolatum* leaf extracts and their AgNP's along with positive control (streptomycin). The data in all figures was measured in at least 5 samples the mean of three replicates S.D. are shown. Statistical analysis was done using ONE WAY ANOVA. \*Denotes significant difference of positive control with test samples at  $P < 0.001$ .



**Fig. 10.** Anticancer activity of *S. jabolatum* leaf extracts AgNP's against MCF-7 cell lines. The data in all figures represents the mean of three replicates S.D. are shown. Data was measured in at least 5 samples, S.D. are shown. Statistical analysis was done using TWO WAY ANOVA with  $P < 0.001$ . \*Denotes significant difference of positive control with test samples at  $P < 0.001$ .



**Fig. 11.** Determining the hemolytic activity of *S. jabolatum* leaf extracts AgNP's towards RBC's of human blood. Data was measured in at least 5 samples, S.D. are shown. Statistical analysis was done using TWO WAY ANOVA  $t$ -test. \*Denotes significant difference of positive control with test samples at  $P < 0.001$ .

cells at different time intervals viz., 5, 30, 60, 120, 180 mints showed  $4.65 \pm 0.11\%$  [52]. Similarly, Shaniba et al. [62] reported that the *Manilkara zapota*-AgNPs on haemolysis in erythrocytes at the  $IC_{50}$  concentration apparently did not cause any damaging effect on human erythrocytes. Decreased and increased haemolysis of 3.1% and 55% were recorded at 6 and 100 µg/mL NPs respectively [41]

## Conclusion

Green synthesis is an eco-friendly procedure which was relatively inexpensive, rapid and easy with no requirement of toxic chemicals. The biosynthesis process was very advantageous over other chemical methods of synthesis AgNPs of leaf extract of *S. jabolatum* AgNPs confirmed successfully served as a reductants and capping agents, analysed by UV-Vis, FTIR, SEM-EDS, and Zeta potential. An absorbance peak in the 335 nm range was observed in UV Visible spectrophotometer. SEM analysis showed that nanoparticles were well-defined with distinctly ordered and skinny sheets. FTIR spectrum revealed that the AgNPs were naturally associated with free hydroxyl groups of phenolic and phenolic compounds. In this study, free hydroxyl functional groups and effective compounds were responsible for the bioreduction process and the stabilization of biosynthesized AgNPs. Zeta potential analysis indicated the presence of more negative charge on the nanoparticles. Biological characterization of AgNPs shows the presence of phenolic and phenolic compounds evaluated by phytochemical screening and these are responsible for 81% antioxidant activity at 25 µg/ml as well as antibacterial activity against gram<sup>+</sup> and gram<sup>-</sup> bacteria similar to streptomycin. Further AgNPs demonstrated cytotoxicity while assays showed a decrease in cell viability of cancerous cell line MCF-7 at 0.0312 µg, besides revealing anti-haemolytic activity at low concentration giving an impression that these nanoparticles are non-haemolytic and nontoxic. All the activities were compared with plant extract and determines the nanoparticles significance in enhancing the activities. Hence due to these properties *S. jabolatum* AgNPs are used in biomedical purpose.

## Declaration of Competing Interest

There are NO conflicts of interests in this paper.

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